

Spatial Mass distributions and Galactic Stability

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The Problem

Via newtonian mechanics and a gaussian integral we expect the velocity distribution of galaxies to look like:
The data shows it follows a curve like:

 center

Whats been Tried

- Newtonian physics

$$F = -\frac{GMm}{r^2}\hat{r}$$

- General Relativistic

$$R^\lambda_{\gamma\alpha}$$

What I am trying

Variable mass:

$$M(r) \propto \sqrt{r}$$

Constant mass:

$$M(r) \propto M$$



Why I am trying it do

- light traveling through plasma

$$\omega^2 = \omega_p^2 + k^2 \iff E^2 = m^2 + p^2$$

- plasma frequency

$$\omega_p^2 = \frac{e^2 n(r)}{\epsilon_0 m}$$

- Since $n = n(r)$, maybe $m = m(r)$

Why I am trying it cont.

Light incident on plasma

- $\Delta E = 0 \Rightarrow \Delta\omega = 0 \Rightarrow -\Delta\omega_p^2 = \Delta k^2$
- In the case $\Delta v < 0 \Rightarrow \Delta\omega_p^2 > 0$
- Since $\omega_p \propto n \Rightarrow \Delta n > 0$

Why I am trying it cont.

- $\Delta n > 0 \Rightarrow n_1 - n_1 \frac{\sin(\theta_1)}{\sin(\theta_2)} > 0$
- $n_1 - n_1 \frac{\sin(\theta_1)}{\sin(\theta_2)} = \frac{n_1}{\sin(\theta_2)} (\sin(\theta_2) - \sin(\theta_1))$
- since $\theta_1, \theta_2 \in [0, \frac{\pi}{2}]$ for $\Delta n > 0$, $\sin(\theta_2) > \sin(\theta_1)$
- for the domain $[0, \frac{\pi}{2}]$ this implies: $\theta_2 > \theta_1$

The process

$$\mathcal{L} = \mu(r) \frac{\dot{r}^2}{2} + \mu(r) \frac{r^2 \dot{\phi}^2}{2} - U(r)$$
$$\frac{\partial \mathcal{L}}{\partial x^i} = \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}^i}$$

where I'm at

Next steps

Ideas for future

Gravitational Lensing

code

```
import pygame  
import numpy
```


Appendix

Bibliography, References, Appendix. etc.

References

- 1.
7. HEIDEGGER, Martin. **Sein und Zeit**. 11. ed. Tübingen: Max Niemeyer Verlag, 1967.



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beginning